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DARHUB Backend | Mobile and operations integration reference

Pilot API Complete Documentation

Registration, profile, work execution, maintenance, curl cookbook, and verified local checks

Base URL: <http://localhost:5678> | Verified: 2026-09-06

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1. Authentication and route matrix

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Authenticated routes use `Authorization: Bearer <access_token>`. Generic authentication validates the JWT; strict pilot authentication also reloads the user and requires `user_type === 3`.

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Security note: several legacy pilot work routes currently have no authentication middleware: order listing, order detail, start day, update work, image upload, end day, and comments. This reflects the live implementation and should be reviewed before production exposure.

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Method	Endpoint	Auth	Purpose
POST	/api/user/pilot-registration	Optional	Create pilot account and documents.
PUT	/api/user/pilot-edit	Bearer	Update profile and related records.
GET	/api/user/pilot/profile?tab=...	Bearer	Read profile tabs.
GET	/api/pilot/work/orders/:pilot_user_id	None	List assigned orders.
POST	/api/pilot/work/confirm-assignment	Strict pilot	Confirm dates.
GET	/api/pilot/work/order-details/:booking_id	None	Read order detail.
POST	/api/pilot/work/start-day	None	Create/update daily log.
POST	/api/pilot/work/update-work	None	Save work data.
POST	/api/pilot/work/upload-field-image/:log_id	Upload only	Upload field image.
POST	/api/pilot/work/end-day	None	Close daily work.
POST	/api/pilot/work/add-comment	None	Add booking comment.
POST/GET/PUT/PATCH/DELETE	/api/pilot/maintenance...	Bearer	Maintenance lifecycle.

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2. Registration

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POST /api/user/pilot-registration uses multipart/form-data. Required: first_name, last_name, email, password, mobile_number. At least one verification flag must be true. Optional: DOB, Aadhar, PAN, UPI, role, JSON address, and JSON bank details.

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Files: profile_image (JPG/JPEG/PNG); identity/certification fields (JPG/JPEG/PNG/PDF); passbook_image (JPG/JPEG/PNG). Each file is limited to 5 MB and each field accepts up to five files.

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```
address=[{"state":"1","district":"2","block":"3","lane1":"Road
1","village":"Aji","pincode":"360003","is_primary":true}]
bank_details=
[{"bank_name":"SBI","acc_holder_name":"Ramesh","acc_number":"12345","ifsc_code":"SBIN0001","is_primary":true
}]
```

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```
{"success":true,"message":"Pilot registered successfully","user_id":123,"token":"<access_token>"}
```

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Common errors: 400 validation/duplicate data, 401 invalid optional creator token, 500 registration failure.

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3. Profile

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GET /api/user/pilot/profile?tab={main|personal|address|bank|documents} returns {success:true,data:{...}}. Main contains user info and assigned/active order counts. Personal contains identity, DOB, Aadhar, PAN, image, and primary address. Address returns all addresses. Bank returns bank and UPI arrays. Documents returns active uploaded document metadata.

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PUT /api/user/pilot-edit accepts multipart data and updates profile, address, bank, UPI, and documents. It updates the logged-in user by default; another user_id requires admin authorization. Supplied address/bank sets replace existing records.

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4. Work execution

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Orders and assignment

GET /api/pilot/work/orders/:pilot_user_id returns {success,orders}, including dates, acreage, price, status, address, log counts, and drone ID. An unassigned ID returns 200 with an empty array.

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POST /api/pilot/work/confirm-assignment requires strict pilot auth:

```
{"booking_id":123,"accepted_dates":["2026-09-07","2026-09-08"]}
```

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Daily work

POST /api/pilot/work/start-day requires booking_id, pilot_user_id, and working_date; optional fields are copilot and drone. It returns 201 for a new log and 200 for an existing log.

```
{ "booking_id":123,"pilot_user_id":45,"co_pilot_user_id":46,"drone_id":7,"working_date":"2026-09-07" }
```

POST /api/pilot/work/update-work requires booking and date, and accepts completed acres, fertilizer, pesticide, water, notes, start time, and end time. The daily log must exist. **POST /api/pilot/work/end-day** uses the same core body. **POST /api/pilot/work/add-comment** requires booking, pilot, and comment. **POST /api/pilot/work/upload-field-image/:log_id** accepts land_image, JPG/JPEG/PNG.

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5. Maintenance

Maintenance routes use generic Bearer auth. **POST /api/pilot/maintenance** submits immediately; **POST /api/pilot/maintenance/draft** saves a draft. Both accept JSON or multipart and require drone_id.

```
{ "drone_id":7,"maintenance_type":"Scheduled Maintenance","task_name":"Routine Inspection","scheduled_date":"2026-09-07","actual_date":"2026-09-07","engineer_id":12,"overall_status":"Good","checklist":[{"component":"Propellers","expected_status":"Good","actual_status":"Good","remarks":"No cracks"}]}
```

If no checklist is supplied, the default eight-component checklist is created. Health scores are Good/Updated 100, Wear & Tear 70, Damaged 30, Needs Replacement 0, averaged across items.

GET /stats returns totals and health statistics. **GET /** supports search, form_status, overall_status, maintenance_type, page, and limit. **GET /:id** returns task/checklist/attachments. **PUT /:id** updates drafts. **PATCH /:id/submit** submits a draft with an overall status. **DELETE /:id** soft-deletes. Attachment routes add or remove files.

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6. Curl cookbook

```
$BASE="http://localhost:5678"
+$TOKEN="<access_token>"
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+curl.exe -i "$BASE/"
curl.exe -i -X POST "$BASE/api/login" -H "Content-Type: application/json" -d
'{"identifiant":"pilot@example.com","password":"your-password"}'
curl.exe -i "$BASE/api/user/pilot/profile?tab=main" -H "Authorization: Bearer $TOKEN"
curl.exe -i "$BASE/api/user/pilot/profile?tab=documents" -H "Authorization: Bearer $TOKEN"
curl.exe -i "$BASE/api/pilot/work/orders/45"
curl.exe -i -X POST "$BASE/api/pilot/work/confirm-assignment" -H "Authorization: Bearer $TOKEN" -H "Content-
Type: application/json" -d '{"booking_id":123,"accepted_dates":["2026-09-07"]}'
curl.exe -i -X POST "$BASE/api/pilot/work/start-day" -H "Content-Type: application/json" -d
'{"booking_id":123,"pilot_user_id":45,"working_date":"2026-09-07"}'
curl.exe -i -X POST "$BASE/api/pilot/work/update-work" -H "Content-Type: application/json" -d
'{"booking_id":123,"working_date":"2026-09-07","completed_acres":8.5,"notes":"Completed"}'
curl.exe -i -X POST "$BASE/api/pilot/work/upload-field-image/789" -F "land_image=@C:\path\field.jpg"
curl.exe -i "$BASE/api/pilot/maintenance/stats" -H "Authorization: Bearer $TOKEN"
curl.exe -i "$BASE/api/pilot/maintenance?page=1&limit=10" -H "Authorization: Bearer $TOKEN"
curl.exe -i -X POST "$BASE/api/pilot/maintenance/draft" -H "Authorization: Bearer $TOKEN" -H "Content-Type:
application/json" -d '{"drone_id":7,"task_name":"Routine Inspection","overall_status":"Good"}'
curl.exe -i -X PATCH "$BASE/api/pilot/maintenance/10/submit" -H "Authorization: Bearer $TOKEN"
curl.exe -i -X POST "$BASE/api/pilot/maintenance/10/attachments" -H "Authorization: Bearer $TOKEN" -F
"attachments=@C:\path\photo.jpg"
```

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7. Verified local checks

Request	Observed
GET /	200, Welcome to DARHUB Backend
Profile without token	401, NO_TOKEN
Maintenance stats without token	401, NO_TOKEN
Confirm assignment without token	401, no token
Orders/999999	200, empty orders
Invalid registration	400, mobile must be exactly 10 digits

Prepared from live route/controller code. Sample IDs and values are illustrative. Curl checks ran against the local backend on 2026-09-06.

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